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An edge node monitoring method for power line and wireless integration

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Abstract

In view of the security threats posed inside and outside the network due to the weaknesses of network protocols and systems, power line and wireless converged communications have security problems in communication connections that appear in environments where heterogeneous networks overlap. In order to cope with the problems of network delay and data transmission security in smart grid, this paper proposes an online monitoring security protection method based on edge nodes, which manages user trust under the premise of not leaking grid user data, and establishes a personalized machine learning model. This method not only guarantees users a secure communication experience, but also ensures users' real-time network connection and content acquisition, and can obtain a fast and personalized user experience.

Keywords: Wireless communication, Power line carrier communication, Fusion, Edge computing, Authentication;

1. Introduction

With the rapid construction of the national smart grid, various emerging services such as advanced monitoring of the distribution network, distributed access of power sources, and power interaction continue to emerge. The security problem of electric power wireless communication network is becoming more and more prominent, and it has attracted the attention of domestic and foreign researchers. Communication connection is a relatively weak link in terms of security, especially in the power line carrier and wireless fusion communication system, there is no tangible physical transmission medium in the wireless network, and it is in an open public space, so the security of

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the wireless network transmission process is more secure severe. Due to the special environment inside and outside the network, power users are exposed to various security threats, such as eavesdropping, camouflage, black hole attacks, and denial of service. With the development of computer application technology and 5G communication networks, new security threats emerge in an endless stream, and it is urgent to explore new security mechanisms and strategies suitable for current wireless communication networks.

At present, many domestic and foreign researchers have conducted research on the wireless communication security of smart grids. The wireless node attacks in the power wireless network include selective forwarding attacks, witch attacks, black hole attacks, malicious data attacks, and energy-consuming attacks. The malicious node in the wireless network can also send forged control frames by changing the parameters of the link layer, so that other neighboring nodes in the network will enter the dormant state according to the reserved time specified in the forged control frame. The transmission channel of the network has been in an idle state all the time, which seriously deteriorates the communication performance of the entire network. Therefore, malicious node monitoring and identity authentication at the MAC layer are essential communication security mechanisms. For wireless transmission security protection technology, some researchers use attack detection methods based on pattern matching and protocol analysis to collect and analyze information through various systems and network resources to find a security technology for security issues. However, this has an impact on the computing power of the system. And storage resources are too demanding. Some researchers use the wireless link layer collision attack detection, but this detection model cannot classify collision attacks in the same layer of communication network, and the confirmation thresholds of different layers of attacks are different, which leads to unsatisfactory detection results. The security of wireless communication in the smart grid requires a reliable and efficient node monitoring method to ensure the security of the grid and enable users to obtain a faster user experience.

Generally, network security mechanisms include active security mechanisms and passive security mechanisms. However, passive security mechanisms are often used in wireless communication network systems, that is, an identity-based security management model, which is used as the authority basis for user access control. This management model is passive and static, so it is not suitable for the environment of power communication network. In this case, we need to establish an active security mechanism. We need to establish a trust relationship for network nodes from the perspective of trust value, so that network nodes can be monitored and combined with historical interaction information. Analysis can dynamically control access, and actively negotiate to achieve the effect of predicting the potential security risks of the system, and transform the trust relationship between nodes into a trust chain, thereby forming a secure cyberspace. In order to solve the above problems, this paper proposes an online monitoring security protection method based on edge nodes to manage the trust of equipment. This method extends the MAC layer security communication mechanism of the wireless channel to the power line carrier channel, determines the comprehensive evaluation value, and calculates the membership matrix according to the signal sent and received in each frame, and then uses the weighted average degree to obtain the evaluation result. Finally, the node with poor communication quality will not be the next hop node. This method effectively resists the threat of malicious packet loss in the process of power line carrier and wireless fusion data transmission.

2. Application Mode of MAC Layer in Power Line and Wireless Convergence Communication

For the access terminal of the smart grid, the integrated communication of power line and wireless can not only greatly improve the reliability of network communication, but also can be an effective supplement to the high-speed communication technology of optical fiber communication. For the power line and wireless fusion terminal, on the MAC layer of the power line and wireless network, a new convergence sublayer is added to process the conversion of data frames, and additional sublayer technology is used to process data frames to ensure that the power data flow between the two networks can be transparently transmitted. As shown in Figure 1, the convergence sublayer uses the data services provided by the power line network MAC layer and the wireless MAC layer to construct the convergence sublayer frame including the MAC layer frame information of the two, thereby completing the MAC layer data frame for the two networks to ensure the transparent transmission of power data between the two networks.

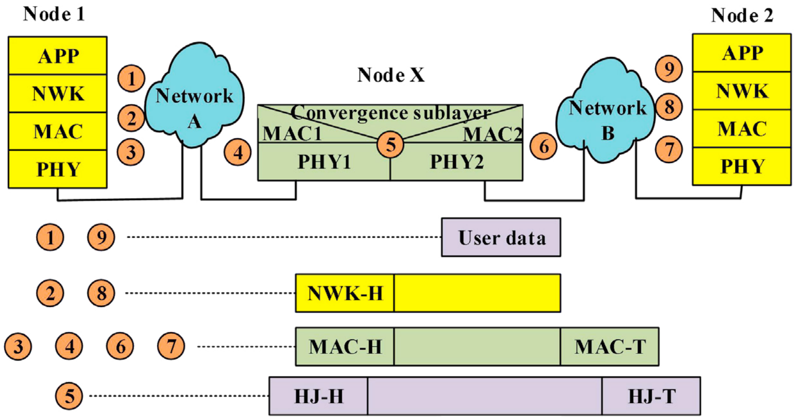

Fig. 1. MAC data frame processing.

Each MAC PDU consists of a 48-bit MAC header, a 48-bit control domain data frame, variable-length payload and cyclic redundancy check bit CRC. General data frame MAC header and message frame MAC header, message frame MAC header PDU does not contain the payload part. The control domain includes a 2-byte connection identifier. The control domain cooperates with the MAC header to complete the MAC layer connection, data frame type, data frame length and other information.

Although the frame headers of the two standard MAC frames used in power line carrier communication and wireless communication are quite different in structure, both include message length and management information. The content that needs to be disclosed will be sent in plain text to ensure that the receiving device can correctly identify and process the content. In order to prevent malicious nodes from malicious packet loss, these nodes must be monitored and authenticated.

3. Design of monitoring model based on edge node

Due to the computational complexity of encryption algorithms, communication terminals must have strong computing capabilities to implement active security mechanisms when transmitting large amounts of power data. However, the computing resources of wireless communication terminals are limited, which results in the use of software to implement cryptographic algorithms. The bottleneck of system processing speed and increase the power consumption of communication nodes. In order to solve the problem of malicious packet loss of MAC layer data due to the low reliability of power line communication, this paper proposes a MAC layer monitoring method based on edge nodes. The forwarding quality of the node is evaluated by the forwarding behavior and deviation of the node, and the historical communication quality is comprehensively considered to evaluate the trust value of the node. Each node transmits the trust value to the cloud for processing. When the edge cloud cannot meet the request of the edge device, it can migrate some tasks and data to the remote cloud for processing. The system architecture is shown in Figure 2:

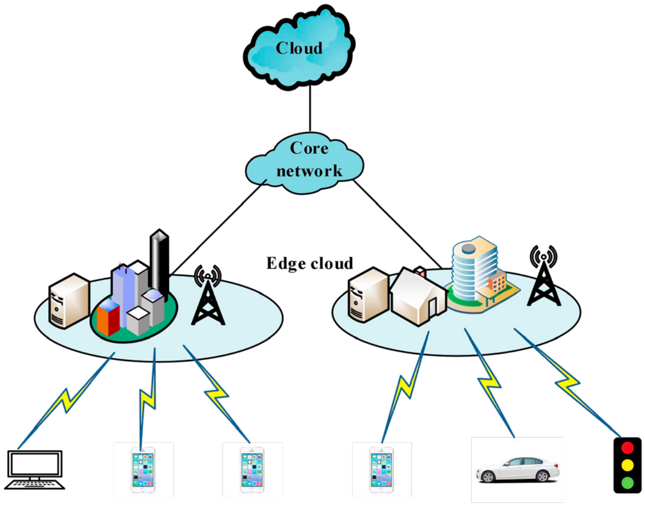

Fig. 2. The working principle of the wireless communication based on edge nodes.

3.1. Trust evaluation mechanism

MAC layer data packets follow the IEEE802.1 protocol, and use OFDM technology to uniformly schedule the power line carrier and wireless spectrum, and send and receive data in a full-duplex manner. Suppose the number of nodes is n , then the set of nodes is $N = \{N_1, N_2, \cdots, N_n\}$, and each node has a unique ID in the network. If node F transmits MAC data packets to node N through node M , then the number of forwarded packets of node M is obtained by monitoring from node M received by node F , then the packet forwarding rate of a certain node can be expressed as:

$$PR_j = \min\left(\frac{NM_j}{NF_i - NF_i \times p_{ij}}\right) \tag{1}$$

Among them, PR_j represents the packet forwarding rate of node j , NM_j represents the number of packets forwarded by node j monitored by node i , NF_i represents the number of packets that need to be forwarded by j , p_{ij} represents the normal packet loss rate caused by the wireless channel and MAC layer between node i and node j .

3.2. Status and monitoring of edge nodes

During the communication process, edge nodes are often in three states: dormant state, monitoring state, and active state. The node in the dormant state is not working and will not consume energy. The node in the monitoring state will continuously perceive the transmission process. Once an abnormal node is found, the monitoring node will immediately turn into an active node and send information about the event to other nodes. The working flow chart of each state is shown in Figure 3:

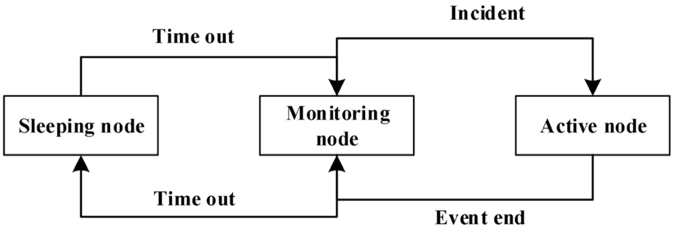

Fig. 3. Node status and transition relationship flowchart.

The energy consumed by a node is also an important condition for measuring the communication quality of a node. The energy deviation of node j can be expressed as:

$$ER_j = \frac{enl_j}{lenergy_j - E_{js}(l)} \tag{2}$$

Among them, enl_j represents the remaining energy of node j, $lenergy_j$ represents the initial energy of node j, and $E_{js}(l)$ represents the energy required for node j to send l data packets. The ratio of the actual remaining energy of the node to the estimated remaining energy is the energy deviation level. Together with the trust value, it is used to comprehensively evaluate the communication quality of the node.

3.3. Discovery and prevention of malicious nodes

The discovery of malicious nodes requires comprehensive consideration of the current communication quality and historical communication quality, that is, the current communication quality is determined by the node's forwarding behavior and energy deviation. The node forwarding behavior represents the integrity of the node, and the node energy deviation represents the communication ability of the node. Communication quality can be expressed as:

$$Q_j = \eta_1 \times PR_j + \eta_2 \times ER_j \tag{3}$$

Among them, the value of Q_j is between 0 and 1, η_1 and η_2 respectively represent the weight of the node's integrity and communication quality, and the value range is also between 0 and 1.

Since the comprehensive communication quality of each node changes dynamically, but the recently entered comprehensive communication quality is more reliable than the historically entered comprehensive communication quality, so when updating the comprehensive communication quality, a weight parameter: time forgetting factor μ needs to be introduced, then the update formula can be expressed as:

$$Q_j = \mu \times Q_{new} + (1-\mu) \times Q_{old} \tag{4}$$

Where Q_{new} is the comprehensive communication quality recently entered by the node, Q_{old} is the comprehensive communication quality entered in the history of the node, $\mu > 0.5$.

When node M monitors that the comprehensive communication quality Q_{MF} of node F is lower than the threshold Q_T set by the system, then node F will be regarded as malicious by the system, and a reward and punishment function will be set. If the feedback function reaches a negative value, the ID of the node is directly added to the blacklist according to the voting result of the neighboring node, and the node no longer participates in any MAC data forwarding task in the network.

4. Design of monitoring model based on edge node

4.1. Discovery and prevention of malicious nodes

To use the wireless public network to access various field terminal equipment of the power system, you need to configure the operator's wireless public network access point name (APN) service, and then access the public network core network through the wireless base station. Then access the main station of the system through the core network gateway in the public network/private network mode. Among them, the public network method is to connect to the system master station via the Internet; the private network method is to access the system master station through the APN access router through the established enterprise private line. APN is the same logo as its fixed Internet domain name assigned by telecom network operators to Internet service providers or other enterprises and institutions. It takes into account the convenience of public networks and the security of private networks. Virtual private network (VPN) business between networks. For enterprise users, the APN service uses IPSec VPN technology to provide efficient remote interconnection services for enterprises and specific industries through the network or established generic route encapsulation (GRE).

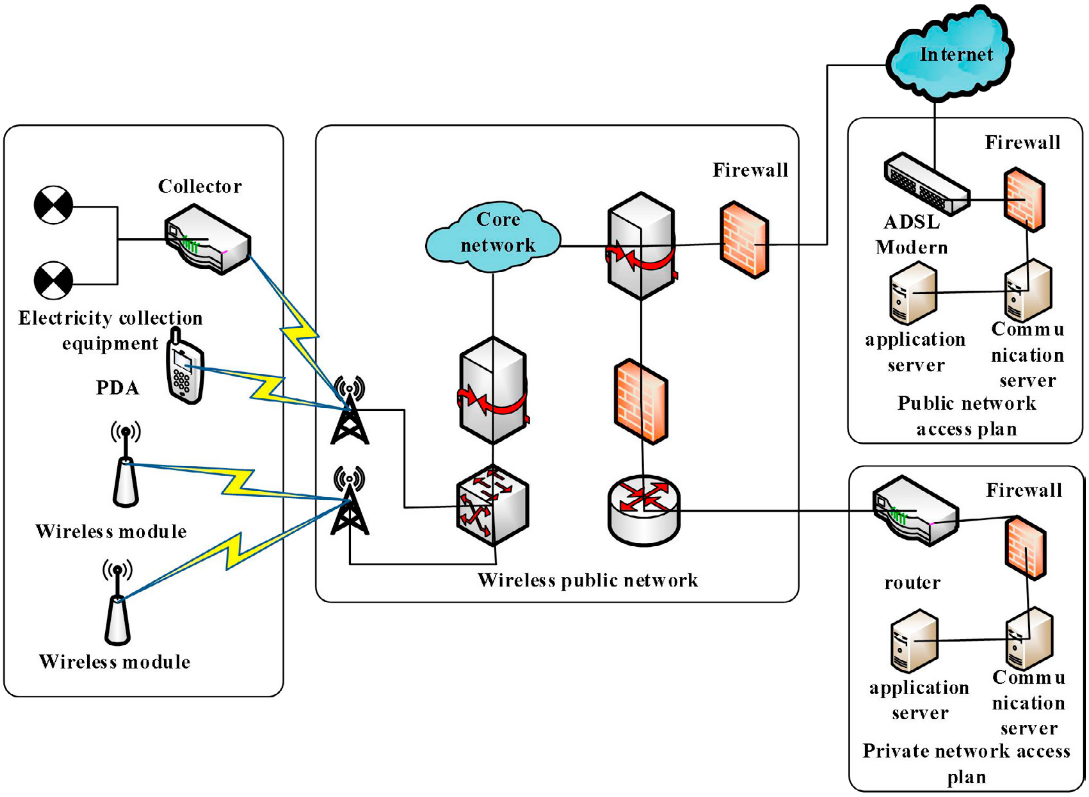

Fig. 4. Application of wireless communication technology in power system.

Node authentication is to maintain a global list of legitimate nodes in the network. In the process of wireless communication, when the system needs to verify the identity of the device, the node needs to show its own certificate to the other party. The node identity authentication based on digital certificate with centralized CA as the core has the advantages of reliability and convenience. However, centralized CA is not suitable for large-scale peer-to-peer networks. On the one hand, there should not be such a central control point in a peer-to-peer network. On the other hand, it also conflicts with the basic idea of full distribution and no management of peer-to-peer networks.

4.2. Identity Authentication of Power Line Carrier Communication Based on Wireless Convergence

The node authentication process in this paper needs to complete the registration of the slave carrier device with the master carrier device, and needs to obtain the connection identifier ID assigned by the master carrier device to the slave carrier device. The node authentication process is initiated from the carrier device. As shown in Figure 5, the complete authentication process includes: the slave carrier device first listens to the line data, waits for the uplink communication channel to occupy an empty gap, and then sends a short message for device authentication; At this time, the main carrier device will receive the device authentication short message, use the downlink channel to occupy the empty gap, and then issue the device to authenticate the permission message, and assign the connection identifier ID ;When receiving the device authentication permission message from the carrier device, waiting for the temporary channel will occupy the logical ID gap, then the sending device can authenticate the information message; The primary carrier device receives the device authentication information message, uses the downlink channel to occupy the empty gap, issues a device authentication permission confirmation message and assigns a connection identifier ID ; if the authentication message is sent over 3 times, the retransmission mechanism is automatically activated.

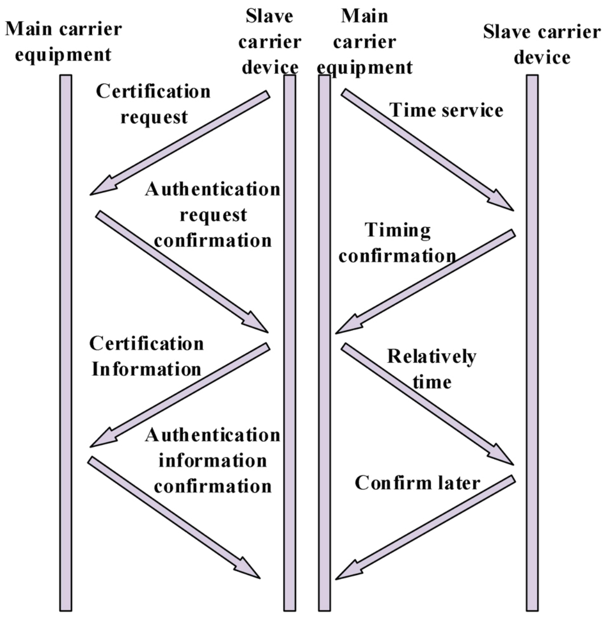

Fig. 5. Node authentication flowchart.

5. Conclusion

Based on the existing research and application of power line carrier communication and wireless communication, this paper analyzes the security of wireless communication connections. According to the attacks in the existing wireless network, an online monitoring security protection method based on edge nodes is proposed. The forwarding behavior and deviation of the node are used to evaluate the forwarding quality of the node, comprehensively consider its communication quality, monitor the nodes whose comprehensive trust value is lower than the threshold, find malicious nodes and add them to the blacklist. Each node transmits the trust value to the cloud for processing. When the edge cloud cannot meet the request of the edge device, part of the tasks and data can be migrated to the remote cloud for processing. It protects users without divulging grid user data, and also ensures that users get a fast and secure communication experience.

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